Calculations report: E2E Final Board

Customer: Test Customer

Revision 02, 2026-10-23

NOT APPROVED: for review only, not for execution.

The design rules are this software's defaults; the company has not confirmed them.

Basis of calculation

- Design current Ib = P / (k U cos phi); breaker In >= Ib, cable Iz >= In (IEC 60364-4-43 §433.1; ABB Electrical Installation Handbook Vol. 2, §2.1-2.3).
- Current-carrying capacity from IEC 60364-5-52 Annex B, corrected for ambient temperature and grouping.
- Voltage drop from the origin, from Schneider EIG Fig. G28 (copper) and ABB §2.2.2 (aluminium), against IEC 60364-5-52 Annex G or the company's limits.
- Fault level at a sub-board through its feeder, IEC 60909-0 (cmax 1.10); breaking capacity at or above it.
- Earth fault disconnection, IEC 60364-4-41 §411.4.4: Zs x Ia <= 0.95 U0, Ia the top of the breaker's instantaneous band (IEC 60898-1).
- Short circuit along the cable, ABB §2.4: the far-end Ikmin trips the breaker at once; k^2S^2 from Table 1 for comparison with the breaker's let-through I^2t.

Board DB1

400 V, 3-phase, TN-S · Ik - kA · Ze - ohm · Incomer C63 4P

Circuit	Ib A	Protection	Cable	L m	dU %	Zs ohm	Zs max	k^2S^2 $(kA)^2s$
Sockets kitchen	9.6	C16 1P	3G2.5 Cu PVC	20	1.5	-	1.372	0.083
Lighting hall	3.8	C200 1P	3G1.5 Cu PVC	20	1	-	0.11	0.03
Split AC	16.8	C20 1P	3G2.5 Cu PVC	20	2.62	-	1.097	0.083

Notes

- Split AC: Ib 16.84 A exceeds the company's 16 A for air_conditioning; rated 20 A instead.
- Group 1 breaker raised to 32 A, 1.6 x the 20 A breaker after it, for discrimination.
- Phase imbalance 77.1 % exceeds the company's 10 %: the loads cannot be spread more evenly as given.
- Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination.
- No fault level given: every breaker's breaking capacity is left to be confirmed.
- Leave 1 spare outgoing ways (20 %).
- Loads without a power factor were taken at cos phi 0.9.

- Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.

- Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.

- Designed for Palestine, under IEC 60364 (adopted by the Palestine Standards Institution): its stated limits apply beneath the company's own settings.

- Circuit rules are this software's defaults, not confirmed by the company's engineers.

- Enclosure: ABB Mini Center compact GDMS332 RX (2CVA250001P0001), 2 rows of 16 modules, H x W x D 498 x 400 x 117 mm, IP41, busbar 200/250 A, 35 kA; 18 modules are left for spare ways (ABB Distribution Boards catalogue (1SKC802027C0201), Mini Center compact, pp. 5, 7). The outgoing terminal strip is not counted.

Findings (1 blocking, 5 warning, 7 info)

[BLOCKING] DB1 / c2: Lighting hall: 200 A protection is above the 17.5 A its 3G1.5 cable carries (method B1); In <= Iz does not hold. (integrity_device_above_cable_capacity; source: ABB Electrical installation handbook

Vol. 2, §2.3 (In <= Iz); Iz from §2.2.1 Tables 4, 5, 8)

[WARNING] DB1 / c3: Split AC: Ib 16.84 A exceeds the company's 16 A for air_conditioning; rated 20 A instead. (above_company_rating; source: policy; engineering confirmation required)

[WARNING] DB1: Phase imbalance 77.1 % exceeds the company's 10 %: the loads cannot be spread more evenly as given. (phase_imbalance; source: company max_phase_imbalance_percent; engineering confirmation required)

[WARNING] DB1: No fault level given: every breaker's breaking capacity is left to be confirmed. (no_fault_level; source: policy; engineering confirmation required)

[WARNING] DB1: Loads without a power factor were taken at cos phi 0.9. (default_power_factor; source: policy; engineering confirmation required)

[WARNING] DB1: Circuit rules are this software's defaults, not confirmed by the company's engineers. (rules_unconfirmed; source: policy; engineering confirmation required)

[INFO] DB1: Group 1 breaker raised to 32 A, 1.6 x the 20 A breaker after it, for discrimination. (discrimination_group_raised; source: policy)

[INFO] DB1: Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination. (discrimination_incomer_raised; source: policy)

[INFO] DB1: Leave 1 spare outgoing ways (20 %). (spare_ways; source: policy)

[INFO] DB1: Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip

(about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables. (discrimination; source: policy)

[INFO] DB1: Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation. (cable_conditions; source: policy)

[INFO] DB1: Designed for Palestine, under IEC 60364 (adopted by the Palestine Standards Institution): its stated limits apply beneath the company's own settings. (market_rules; source: policy)

[INFO] DB1: Enclosure: ABB Mini Center compact GDMS332 RX (2CVA250001P0001), 2 rows of 16 modules, H x W x D 498 x 400 x 117 mm, IP41, busbar 200/250 A, 35 kA; 18 modules are left for spare ways (ABB Distribution Boards

catalogue (1SKC802027C0201), Mini Center compact, pp. 5, 7). The outgoing terminal strip is not counted. (enclosure_selected; source: policy)

Prepared	Checked	Review approved